

Smart Energy Active Learning Pilot

23 April 2026

Purpose: a short project overview in the form of answers to the main questions that came up while reviewing the dashboard and the Active Learning demonstrator.

This document is not intended as a formal report, but rather as a quick yet sufficiently complete overview of:

- what the project currently is,
- what is already clear,
- what our part of the system does,
- and what is still open or requires confirmation

1 What is this project at the moment?

Based on the current understanding, the project is not a single application, but an integrated pilot composed of several parts:

- the HumAIne dashboard on the Ubitech side,
- MinIO as the shared storage for data, models, and metrics,
- the retraining API on our side (IJS, Atena),
- and a model for classifying the security state of the power grid.

The main idea of the pilot is:

1. the dashboard displays model predictions and uncertainty,
2. the user participates in a human-in-the-loop workflow on the dashboard,
3. the dashboard prepares data to extend the training set,
4. clicking **Retrain** triggers our API,
5. our API retrains the model and stores the new model and metrics in MinIO.

So, the dashboard is not only a visualization layer, and our API is not the entire Active Learning system. Each part covers its own part of the pipeline.

2 What are we classifying?

We are classifying the **security state of the power grid** for a given operating point.

The model receives as input:

- loads (`load_*`),

- generator outputs (**gen_***),
- static generator outputs (**sgen_***),
- and several aggregated grid metrics, such as maximum line loadings and voltages.

The current reference dataset on our side contains:

- **8769 rows**
- **273 columns**
- including:
 - 20 **load_*** features,
 - 135 **gen_*** features,
 - 110 **sgen_*** features,
 - 6 aggregated metrics for base-case and contingency state.

The target label is binary:

- **secure**
- **insecure**

3 What does **secure** and **insecure** mean?

This is the most important part, because it is also the least intuitive thing currently visible from the dashboard.

According to the code on our side, **secure/insecure** means the following:

A state is **secure** if:

- the base-case power flow succeeds,
- there are no constraint violations,
- and all checked N-1 contingencies also pass successfully.

A state is **insecure** if any of the following happens:

- the power flow does not converge,
- a line exceeds 100% loading,
- the voltage at a bus drops below 0.9 pu,
- the voltage exceeds 1.1 pu,
- or any N-1 outage of a line or generator causes such a problem.

Therefore, **insecure** here does **not** directly mean:

- total blackout,
- danger to the end user,
- or some vague general “bad” status.

It means more precisely:

a state that does not pass the security check under the applied N-1 criterion and simulator.

This is very important, because without this context the user cannot interpret the prediction properly.

4 What does “probability of insecurity” mean?

Based on the current understanding, it means:

the model-estimated probability that a given operating point belongs to the **insecure** class.

More concretely:

- the current model is a Random Forest classifier,
- the model outputs a class probability,
- the dashboard then displays the probability for the **insecure** class.

Again, this is not the same as:

- probability of blackout,
- probability of total grid failure,
- or probability of a harmful event in a broader sense.

If we want this to be understandable to users, it would be better to describe it more precisely in the documentation or paper, for example as:

- probability of violating the current N-1 security criterion,
- probability of insecure operating state under the used simulator criterion.

5 Where is the Active Learning part here?

In the current architecture, Active Learning is not centralized in our API, but split between the dashboard and the retraining component.

5.1 What appears to be the Active Learning part

The Active Learning part is:

- identification of low-confidence or uncertain cases,
- selection of cases that we want to add for further learning,
- and retraining on the extended dataset.

5.2 Where that logic is located

Based on the current implementation:

- **Ubitech / dashboard side:** sample selection, uncertainty-based logic, adding new rows
- **our side:** retraining on updated data

This means that our retraining API is not, by itself, an “Active Learning selector.” It does not choose samples; it only retrains the model on the data prepared by the dashboard.

6 Where is the human-in-the-loop part?

This is still partly open and should be explicitly aligned with Ubitech.

Based on the communication so far, the idea seems to be:

- the dashboard highlights uncertain or interesting points,
- the user reviews them,
- these points are used to augment the dataset,
- and retraining is then triggered.

What is still not fully clear is the following:

- whether the human **actually labels** these points manually,
- whether the human only **selects** them,
- whether the human only **confirms** them,
- or whether the human is mainly the person who triggers the workflow and retraining.

This is a crucial difference.

If the human directly provides the label, then this is a classical HITL / expert labeling scenario. If the human only selects interesting points and the final label still comes from the simulator or another process, then this is a different type of HITL story.

For the paper and for system interpretation, we need to know exactly which of these explanations is correct.

7 What exactly does the Retrain button do?

According to the current implementation on our side, the **Retrain** button means the following:

1. the dashboard calls our FastAPI endpoint `POST /retrain`,
2. our API downloads the latest file containing appended rows from MinIO,
3. it merges that file locally with the base dataset,
4. it retrains the Random Forest model,
5. it computes metrics,
6. it uploads the new model and metrics back to MinIO,
7. and it returns the artifact locations and results.

Important:

- retraining does not mean simply “recompute the graph,”
- it is not an inference call,
- and it does not mean that the Active Learning logic itself is executed on our side.

It is the retraining of the model on an expanded training set.

8 Who is the system intended for?

This is one of the main open questions.

At first glance, the system could be intended for:

- a system operator,
- a domain expert,
- a researcher,
- or a demonstrator user within the pilot project.

At the moment, it is not completely clear from the dashboard which of these profiles is the main one.

This matters because it determines:

- how much ML knowledge we expect from the user,
- how uncertainty should be explained,
- how `secure/insecure` should be explained,
- and whether it even makes sense for that user to click **Retrain**.

If the target person is a real operator, then the workflow should be explained in operational and domain terms. If the target person is a researcher or demonstrator, then a more technical explanation may be acceptable.

9 What is currently confirmed and sufficiently clear?

The following points are, based on the current review, fairly well confirmed:

- there is a working retraining API on our side,
- the API exposes at least `GET /health` and `POST /retrain`,
- the dashboard and our API are already integrated,
- the current model is a Random Forest classifier,
- the classification uses the labels `secure / insecure`,
- our retraining component uses the local base dataset plus the latest appended rows,
- the retraining results are uploaded to MinIO as model + metrics.

10 What is still not sufficiently clear?

The following points are still open and should be confirmed mainly with Ubitech:

- who the main system user is,
- whether the human only selects a sample or also labels it,
- where the ground truth for newly added samples comes from,
- what exactly the user sees when evaluating uncertain points,
- whether explainability is really part of the current workflow or more of a future direction,

- how exactly the Active Learning strategy is presented in the dashboard,
- and how the meaning of `probability of insecurity` should best be explained to the user.

11 What do the screenshots clarify?

The reviewed screenshots do help clarify the platform structure and the role of some modules, even if they do not yet answer all conceptual questions.

The screenshots clarify the following points:

- the platform is clearly split into **Card Feed**, **Visualization**, **Digital Twin**, and **Operator Agent**,
- there is a dedicated **N-1 Day-Ahead Security Classification** view,
- the **Re-training** action is currently attached to the classification dashboard card,
- the **Digital Twin** appears as a separate simulation-oriented module with date, time, line-outage selection, and a simulation trigger,
- the **Operator Agent** appears as a session-based chat/tool interface that can retrieve notifications, fetch details, inspect topology, run simulations, and trigger forecasting or classification jobs,
- the **Visualization** section confirms that the platform is broader than only Active Learning, since it also includes day-ahead total load and electricity price views.

At the same time, the screenshots still do **not** fully clarify:

- what exactly `secure` and `insecure` mean for the end user,
- what exactly the displayed probability should be interpreted as,
- where the actual sample-selection step happens,
- whether the human only selects points or also labels them,
- and what concrete explainability support is currently available beyond UI placeholders.

So, the screenshots are very useful for understanding the **platform layout and workflow entry points**, but they do not yet fully resolve the **conceptual interpretation of the Active Learning / HITL logic**.

12 Main questions

1. Who is the primary user of this workflow?

The primary user of the workflow is the transmission system operator. The system is designed as an operator-support tool for inspecting day-ahead security assessment results, triggering additional Digital Twin simulations, reviewing uncertain cases, and deciding whether retraining should be initiated. During the pilot phase, technical users and researchers may also interact with the workflow for testing and demonstration purposes, but the intended operational user remains the system operator.

2. Does the user manually label the sample or only select it?

In the current workflow, the user does not manually assign the secure/insecure label. The user selects or confirms uncertain or informative operating points that should be further evaluated. The actual ground-truth label is then produced by the Digital Twin through N-1

simulation. Therefore, the human-in-the-loop role is mainly sample selection, validation, and workflow control, rather than direct manual labeling.

3. What is the exact source of ground truth for newly added samples?

The ground truth for newly added samples comes from the Digital Twin simulation. For each selected operating point, the Digital Twin runs the base-case AC power flow and the relevant N-1 contingency simulations. A sample is labeled secure if the base case and all tested contingencies converge and remain within the defined operational limits. A sample is labeled insecure if the power flow does not converge, if any line exceeds 100% loading, or if any bus voltage falls outside the accepted voltage range of [0.90, 1.10] pu.

4. What does the user actually see when evaluating an uncertain sample?

The user sees the operating point under evaluation, the predicted security class, and the model-estimated probability of insecurity. The user can inspect the timestamp and relevant prediction information, and can trigger a Digital Twin simulation for the uncertain case. After the simulation is executed, the user can review the resulting secure/insecure label, line loading values, bus voltage values, and any detected violations. This allows the user to understand whether the uncertainty corresponds to a real N-1 security issue.

5. Which Active Learning strategies are currently really active in the dashboard?

The dashboard currently supports an uncertainty-driven workflow based on the model-estimated probability of insecurity. In practice, uncertain samples are those for which the classifier has lower confidence or probabilities closer to the decision boundary. Other Active Learning strategies, such as entropy sampling, margin sampling, and random sampling, have been used in the experimental evaluation, but they should not be described as active dashboard options unless they are explicitly exposed in the user interface.

6. How should we best describe **insecure state** to the user?

An insecure state should be described as an operating point that does not satisfy the implemented steady-state N-1 security criterion in the Digital Twin. More specifically, it is a state where the base case or at least one tested contingency leads to power-flow non-convergence, line loading above the allowed threshold, or bus voltage outside the accepted range. It should not be described as a direct probability of blackout or total grid failure.

7. Is explainability an actually implemented part of the current workflow?

Explainability is currently implemented for the day-ahead total load forecasting workflow. XAI outputs such as feature importances, forecast contributions, detailed importances, partial dependence plots, residuals, and weak-spot analysis are generated and visualized in the dashboard per timestep. For the security classification / Active Learning workflow, explainability should only be claimed if classifier-specific explanations are also implemented and shown in the interface. Otherwise, XAI should be described as implemented for forecasting and complementary to the Active Learning workflow.

8. How do we want to present this work in the paper: as an integration pilot or as an operator-aware Active Learning story?

The work should be presented as an operator-aware Active Learning workflow for Digital Twin-based N-1 security assessment. The key contribution is not only the technical integration of dashboard, MinIO, retraining API, and classifier, but also the human-in-the-loop process in which the operator reviews uncertain grid states, triggers additional Digital Twin simulations, and initiates retraining on the expanded dataset. If the emphasis is on

the broader platform, it can also be described as a Human-AI integration pilot, but the strongest research framing is the operator-aware Active Learning story.

Conclusion

The technical integration appears, based on the current understanding, to be largely in place. The biggest open issue is no longer “does the system work,” but rather “how should it be explained correctly.”

For a complete understanding of the project, we therefore need to keep two things in view at the same time:

- technically: dashboard + MinIO + retraining API + model,
- conceptually: who the human in the loop is, what exactly that person does, and what the shown result actually means.

Once these questions are further confirmed, it will be much easier to present the project clearly to others as well.

A Referenced Platform Screenshots

The current document includes the following screenshots from `figures/platform`.

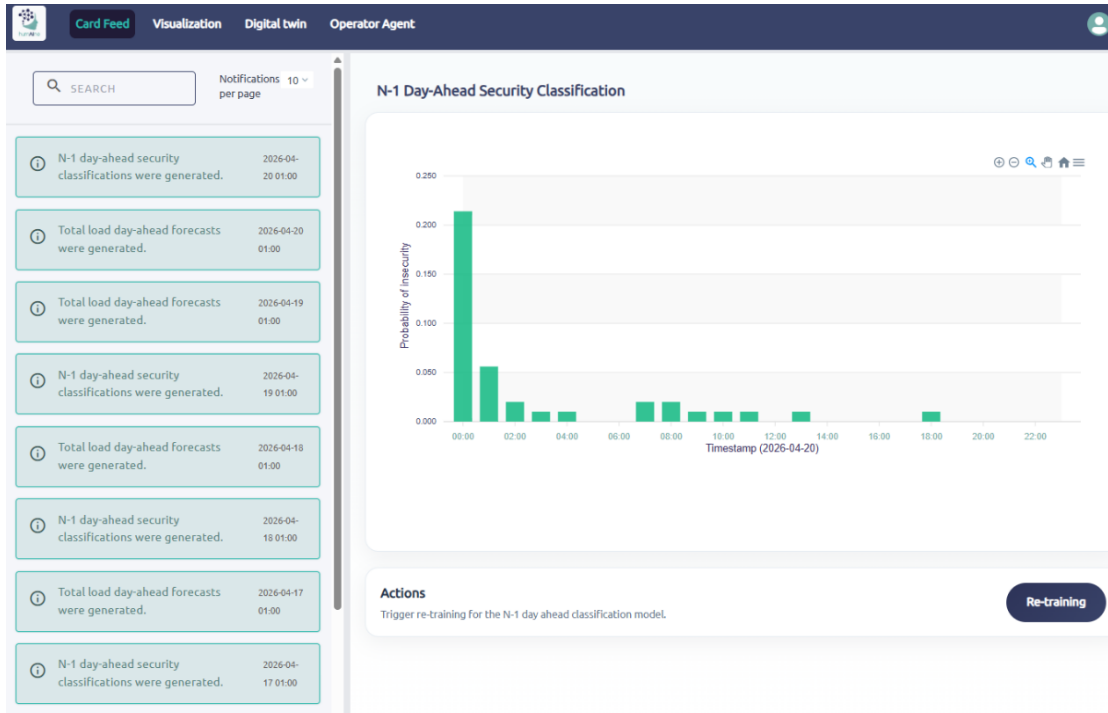


Figure 1: N-1 day-ahead security classification dashboard. This view shows the dedicated classification result page and makes it clear that the retraining action is attached to the classification workflow.

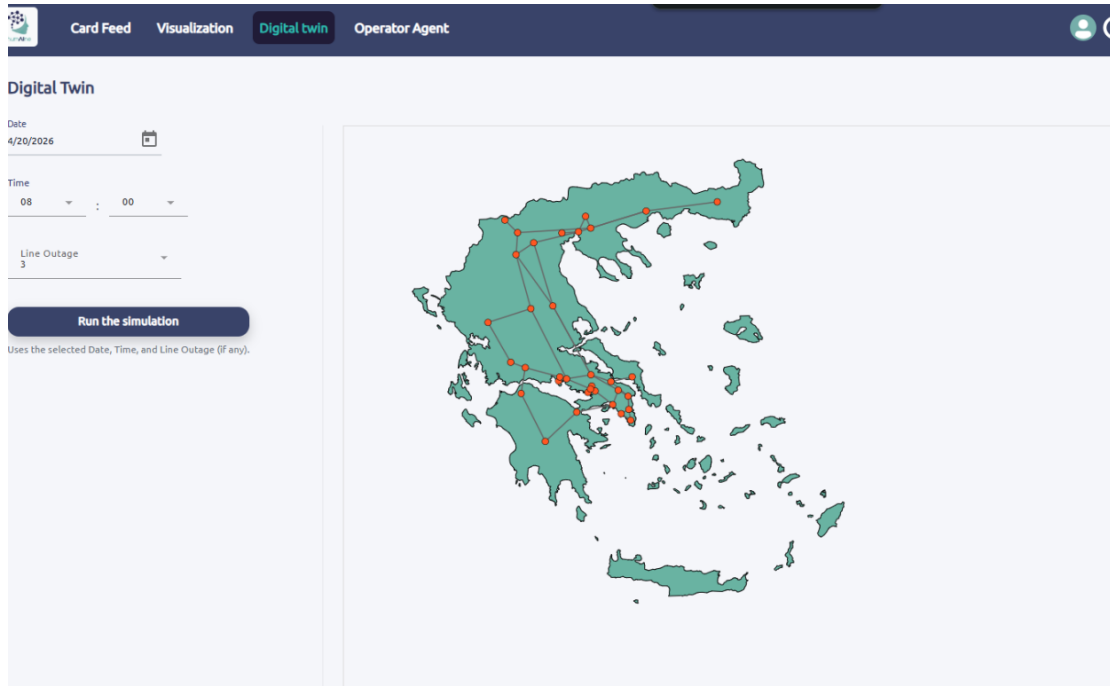


Figure 2: Digital Twin simulation interface. This view clarifies that the digital twin is presented as a separate simulation module with date, time, line outage selection, and a simulation trigger.

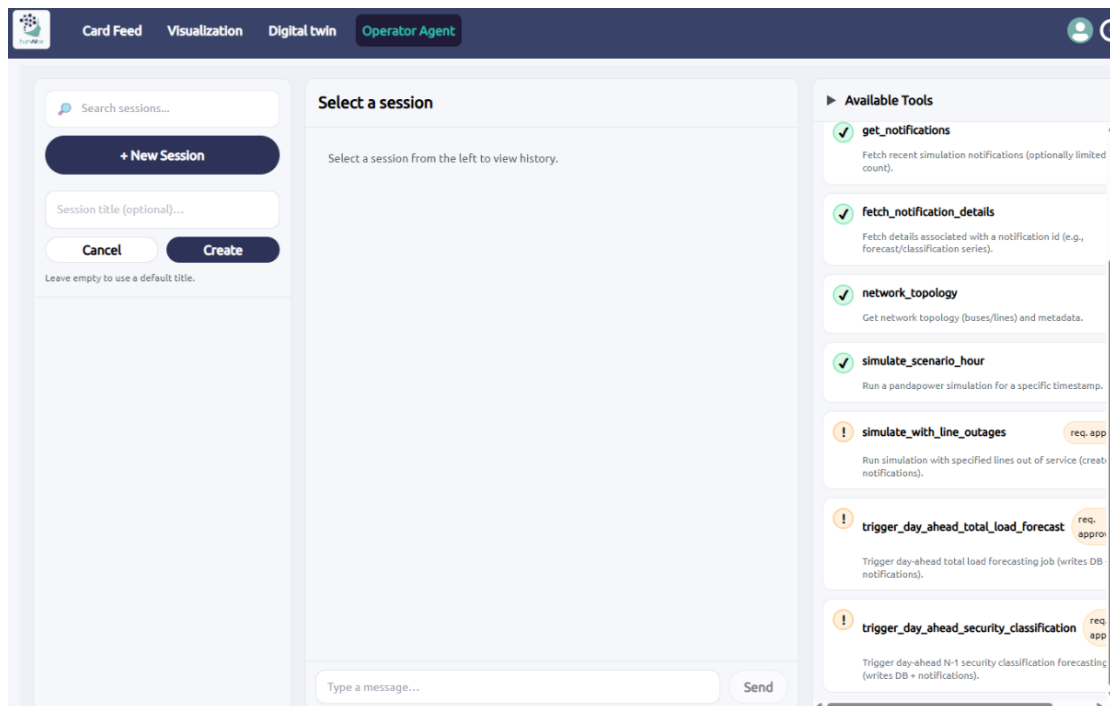


Figure 3: Operator Agent session management interface. This screenshot shows the session-oriented chat structure and confirms that the operator agent is organized around user sessions.

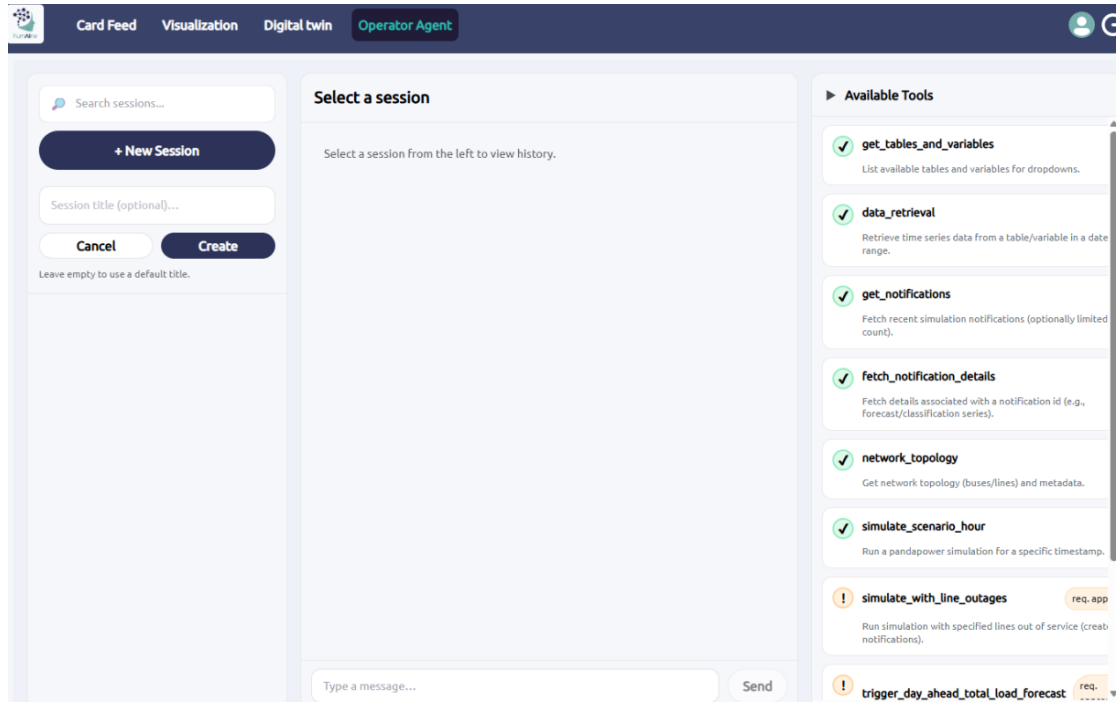


Figure 4: Operator Agent interface with available tools. This screenshot clarifies that the operator agent can call multiple platform tools, including notifications, topology retrieval, simulation, and triggering forecast or classification jobs.

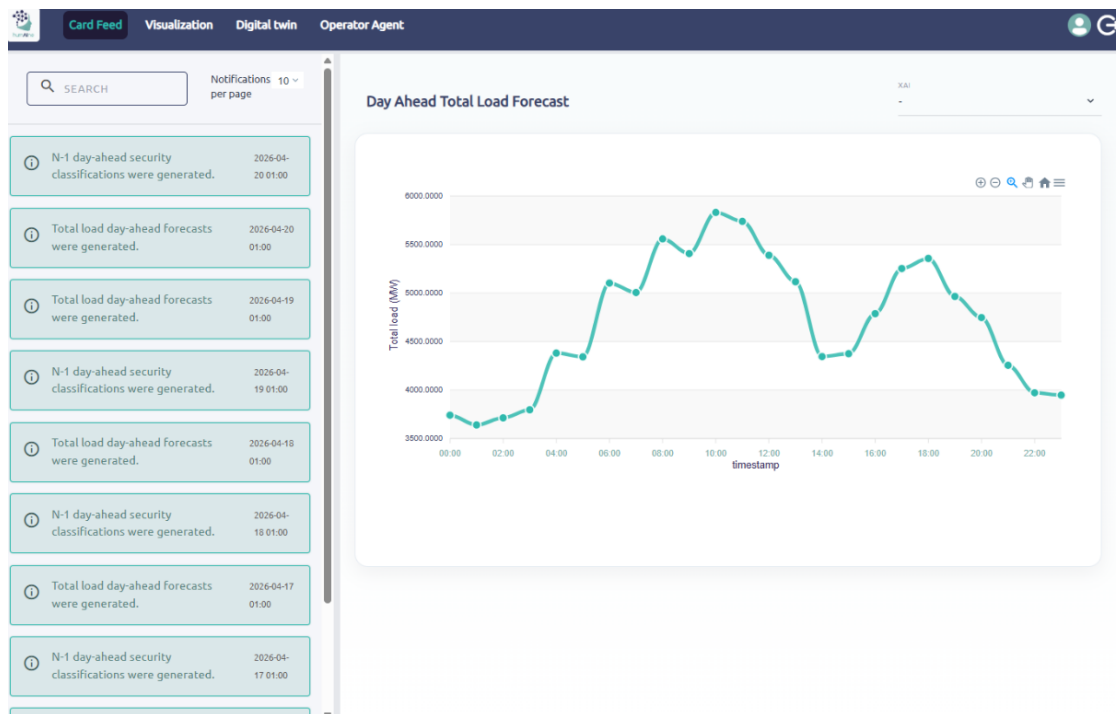


Figure 5: Day-ahead total load forecast visualization. This screenshot shows that the broader platform includes standard forecasting and visualization modules in addition to the Active Learning-related components.

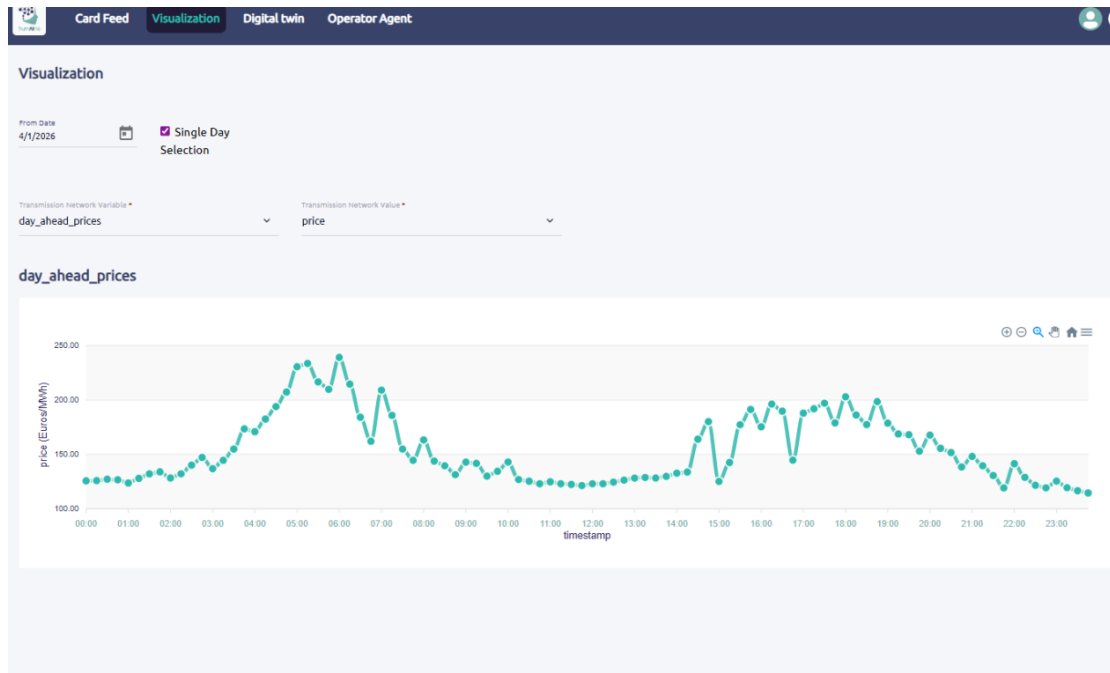


Figure 6: Day-ahead electricity prices visualization. This screenshot further confirms that the platform contains general energy analytics views and is not limited only to the security classification workflow.